DPP DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	CHEST TUBE INSERTION, CARE AND REMOVAL
DEPARTMENT:	INTENSIVE CARE UNIT
APPLIES TO:	ALL MEDICAL AND NURSING STAFF

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1. PURPOSE

1.1. To formulate standardized criteria for inserting, caring and removing chest drainage.

2. DEFINITION & ABBREVIATION

2.1. **Chest tube** - a hollow, flexible tube inserted into the space between the lung and the chest wall (the pleural space) to drain air, fluid, blood, or other material from the chest.

2.2. **Pneumothorax** is an accumulation of air in the pleural space.

2.3. **Pleural Effusion** is abnormal buildup of excess fluid in the pleural space.

2.4. **Hemothorax** - the accumulation of blood in the pleural cavity.

2.5. **Oscillation** - called tidaling, is the normal fluctuation of the water level in the water seal chamber that reflects the changes in intrathoracic pressure with breathing. Oscillation is a positive sign that the drain is functioning correctly and that the lung is expanding.

2.6. **MCC** – Medica Cloud Care System (HIS – Hospital Information System).

3. RESPONSIBILITY

3.1. Physician

3.1.1. Assessing patient's condition before and after chest drainage insertion and removal.

3.1.2. Entering the order for the procedure and medication needed in the MCC.

3.2. Head Nurse

3.2.1. Unit staff are knowledgeable in assisting for insertion, caring and removal of chest drainage.

3.2.2. All needed equipment are available in the unit for emergency cases.

3.3. Charge Nurses

3.3.1. Delegate the procedure to a competent nurse and help the staff nurse throughout the procedure.

3.3.2. Double check physicians' orders and medications.

3.4. Staff nurse

3.4.1. Carrying out physician's orders.

3.4.2. Preparing needed Equipment and assisting physician.

3.4.3. Frequently reassess patient's condition.

4. POLICY

4.1. Insertion and removal of chest drainage shall be performed by a surgeon with assistance of ICU Nurse.

4.2. Aseptic technique shall be observed throughout the procedure.

4.3. Chest tube shall be monitored for air leak every hour and pleurovac chamber will be monitored for fluctuation every hour.

4.4. Rubber tipped clamps should be ready at bedside.

4.5. Chest tube must be clamped for several hours before removal by physician with order for that.

4.6. Chest tube set shall be readily available for any emergency cares.

5. PROCEDURE

5.1. Chest Tube Insertion

5.1.1. Preparatory Phase

5.1.1.1. Assess patient for pneumothorax, hemothorax, presence of respiratory distress.

5.1.1.2. Obtain a chest x-ray, ultrasound or fluoroscopic as ordered.

5.1.1.3. Surgeon will obtain informed consent after explanation of the procedure and ICU nurse will sign as witness. (Refer to APPH-QPS-0008 Informed Consent).

5.1.1.4. Reassure the patient and explain briefly the procedure.

5.1.1.5. Verify right patient and right location / procedure.

5.1.1.6. Prepare needed equipment and assemble drainage system as manufacturer instruction.

5.1.1.7. Monitor vital signs and give Analgesia as ordered.

5.1.1.8. Position the patient according to the insertion site.

5.1.1.9. Using standard precaution, the skin is prepared, anesthetized using local Anesthetic and draped. (Refer to MPP-IPC-0001 IPC Guidelines in Standard Precautions).

5.1.2. Performance Phase

5.1.2.1. Needle or intra-catheter technique:

5.1.2.2. Using aseptic technique, a larger needle is used to infiltrate the subcutaneous tissue, intercostal muscles and parietal pleura.

5.1.2.3. The intra-catheter is inserted through the needle into the pleural space. The needle is removed and the catheter is pushed several centimeters into the pleural space.

5.1.2.4. The catheter is taped to the skin; sutured to the chest wall and covered with a dressing.

5.1.2.5. The catheter is attached to a connector / tubing and attached to a drainage system (underwater seal or commercial system) and all connections taped.

5.1.2.6. Trocar technique:

5.1.2.6.1. Using aseptic technique, a small incision is made over the prepared,

anesthetized site. Blunt dissection (with a hemostat) through the muscle planes in the interspaces to the parietal pleura is performed.

5.1.2.6.2. The trocar is directed using pointed rod for ease of insertion into the pleural space through the incision, the metallic trocar is removed, and a chest tube is inserted into the pleural space and connected to a drainage system.

5.1.2.6.3. The tube is sutured in place and covered with sterile dressing to prevent dislodgment.

5.1.2.6.4. The tube is attached to a connector tube and to the system. All connections are taped.

5.1.2.6.5. Clamps are removed from the chest tube once connected to the drainage system. Chest tubes open to air at the time of insertion will result in pneumothorax.

5.1.2.6.6. Connect to suction port and set the vacuum pressure if ordered.

5.1.2.6.7. Keep the draining system below the level of the chest in upright position and observe for oscillation.

5.1.2.6.8. Monitor vital signs throughout the procedure.

5.1.2.6.9. Verify the correct placement of the chest tube by doing x ray.

5.1.2.6.10. Document time of chest tube insertion, suctioning pressure if connected, name of physician performing the procedure, time of chest x-ray done, initial amount of drainage and patient tolerance to procedure in the progress notes (MCC).

5.1.3. **Follow-up phase**

5.1.3.1. Observe the drainage system for blood and air. Observe for fluctuation in the tube on respiration.

5.1.3.2. Secure a follow-up chest x-ray as ordered by physician.

5.1.3.3. Monitor vital signs every hour or as per physician's order, observing for signs of dyspnea, restlessness, instability and fever.

5.1.3.4. Mark the drainage from the collection chamber with tape/ marker on the outside of the drainage Unit.

5.1.3.5. Do not elevate drainage set above level of chest and avoid fall down of the system during patient's turning.

5.1.3.6. Assess for bleeding infection, leakage of air and fluid around the tube.

5.1.3.7. Assess for oozing or bleeding from dressing site.

5.1.3.8. Maintain integrity of the chest drainage system.

5.1.3.9. **Chest Tube Removal**

5.1.3.9.1. The physician will decide to remove the chest tube according to clinical assessment and chest X-ray.

5.1.3.9.2. Identify the correct patient using two identifiers (Refer to APP-QPS-0016 Patient Identification).

5.1.3.9.3. Explain the procedure to the patient.

5.1.3.9.4. Position the patient according to physician's order.

5.1.3.9.5. Assemble all equipment in the bed side.

5.1.3.9.6. Pre-medicate with analgesics 30 minutes before the procedure according to doctors order.

5.1.3.9.7. Wash hands and follow aseptic technique.

5.1.3.9.8. The physician will clamp the tube.

5.1.3.9.9. Check the initial vital signs.

5.1.3.9.10. Remove present dressing and cleanse the site using antiseptic solution.

5.1.3.9.11. The physician removes the sutures holding chest tube in place.

5.1.3.9.12. The physician quickly pulls out the tube while asking patients to take deep breathing.

5.1.3.9.13. Take 4x4 sterile gauze and place over the site, hold dressing firmly in place and secure with adhesive plaster immediacy.

5.1.3.9.14. Discard all the supplies in the respective bags as per infection control policy.

5.1.3.9.15. Have chest x-ray done after the procedure.

5.1.3.9.16. Observe the patient, check the viral signs after the procedure and check the dressing site for oozing.

5.1.3.9.17. Wash hands thoroughly. (Refer to APPH-IPC-0005 Guidelines in Hand Hygiene).

5.1.3.9.18. Document the procedure, any events during the procedure, patient's tolerance to the procedure and the vital signs in the progress notes (MCC).

6. MEASUREMENT

6.1. 100% of the medical and nursing staff are well oriented and adheres to this policy.

6.2. 100% of patients are assessed by the Physician before Chest Tube insertion and removal.

6.3. 100% of the procedure will be performed using Standard precaution.

7. REFERENCES /CROSS- REFERENCES

7.1. MOH Pocket Manual in Critical Care 2020

7.2. https://medlineplus.gov/ency/article/002947.htm

7.3. APPH-QPS-0008 Informed Consent

7.4. MPP-IPC-0001 IPC Guidelines in Standard Precautions

7.5. APP-QPS-0016 Patient Identification

7.6. APPH-IPC-0005 Guidelines in Hand Hygiene

8. ATTACHMENT

8.1. None

9. REVISION CLAUSE

9.1. N/A